

COMPASSIQ - AI POWERED TICKET MANAGEMENT SYSTEM



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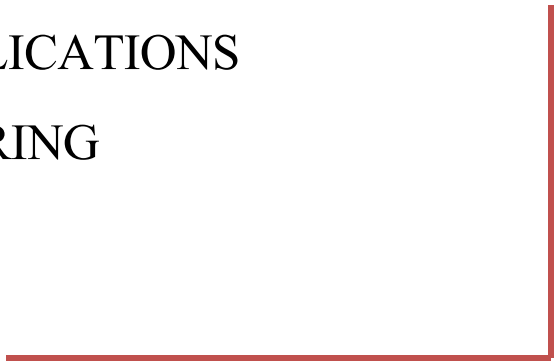


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INTRODUCTION

- Customer support is one of the most important services offered by product-based companies.
- Every day, organizations receive numerous customer complaints related to technical issues, billing, account support, and general inquiries.
- Processing these tickets manually is time-consuming and may lead to incorrect ticket routing and delayed responses.
- CompassIQ is an AI-powered customer support ticket intelligence system designed for a consumer electronics company.

INTRODUCTION

- The application uses Natural Language Processing (NLP) and Machine Learning (ML) to analyze customer complaints, predict the issue category and priority, retrieve the Top-3 similar historical tickets using TF-IDF and Cosine Similarity, and automatically route tickets to the appropriate department.
- The system aims to improve support efficiency, reduce manual effort, and provide a faster and more organized customer support experience.

OBJECTIVES

- To automate customer support ticket management.
- To automatically classify customer complaints into predefined support categories.
- To predict the priority level of each support ticket.
- To automatically route tickets to the appropriate department.
- To retrieve Top-3 similar historical complaints to provide useful context to support staff.
- To provide AI-based insights to support staff using similar historical tickets.
- To reduce manual effort and response time.
- To improve customer satisfaction through faster support.

EXISTING SYSTEM

- Manual ticket categorization leads to delays and inconsistencies.
- Incorrect department assignment increases resolution time.
- Slow response due to manual processing of support requests.
- High workload for support staff in handling repetitive tasks.
- Reduced customer satisfaction because of delayed and inaccurate support

PROPOSED SYSTEM

- Accepts customer complaints through an online portal.
- Preprocesses complaint text using NLP techniques.
- Converts complaint text into TF-IDF feature vectors.
- Predicts ticket category using Logistic Regression.
- Predicts ticket priority using Logistic Regression.
- Retrieves Top-3 similar historical tickets using TF-IDF and Cosine Similarity.
- Automatically routes tickets to the appropriate department.
- Provides separate dashboards for Customers, Departments, and Admin.
- Allows departments to respond, update status, and record resolution details.
- Reduces manual effort and improves support efficiency

MOTIVATIONS

- Consumer electronics companies receive a large number of support tickets daily.
- Manual ticket handling is repetitive, time-consuming, and error-prone.
- AI automates ticket classification and priority prediction.
- Historical tickets can be used to identify similar customer complaints and provide useful context to support staff.
- Improves operational efficiency and reduces manual workload.
- Enables faster and more consistent customer support.

FUNCTIONALITIES

Customer Module

- Secure user registration and login.
- Submit support tickets with issue details.
- Track ticket status.
- View responses from the assigned department.
- Provide feedback and ratings after resolution.

AI Intelligence Module

- Clean and preprocess complaint text using NLP.
- Convert complaint text into TF-IDF feature vectors.
- Predict ticket category and priority using Logistic Regression.
- Retrieve Top-3 similar historical complaints using Cosine Similarity.
- Automatically determine the appropriate department based on predicted category.

FUNCTIONALITIES

Department Module

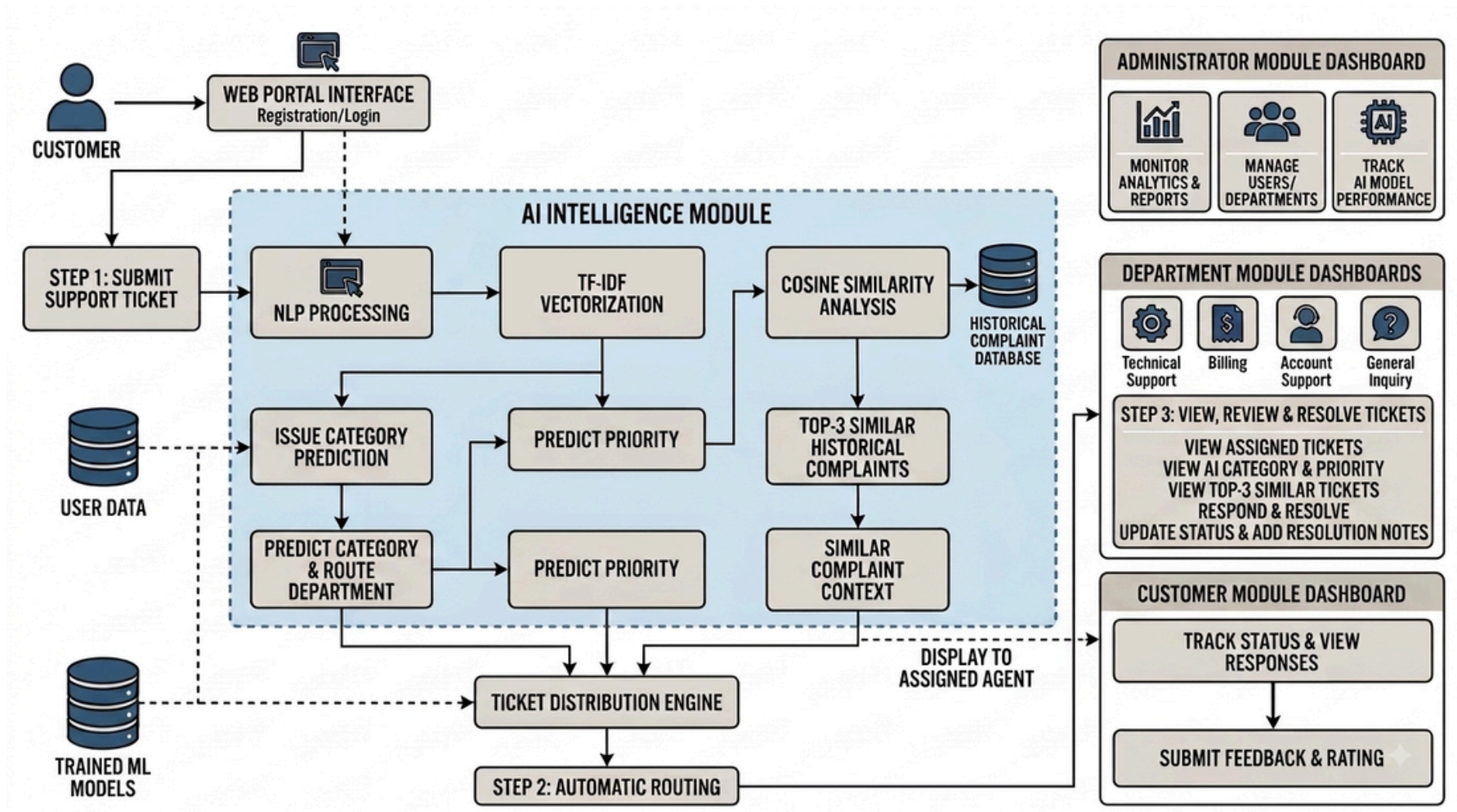
- Separate dashboards for Technical Support, Billing, Account Support, and General Inquiry.
- View tickets automatically assigned to the department.
- View AI-predicted category and priority.
- View Top-3 similar historical complaints.
- Respond to customer queries.
- Update ticket status.
- Add final resolution notes before closing tickets.

FUNCTIONALITIES

Administrator Module

- Manage users and departments.
- View overall ticket statistics.
- View category and priority distribution.
- Monitor ticket status and department performance.
- Monitor AI model performance.
- Generate basic system reports

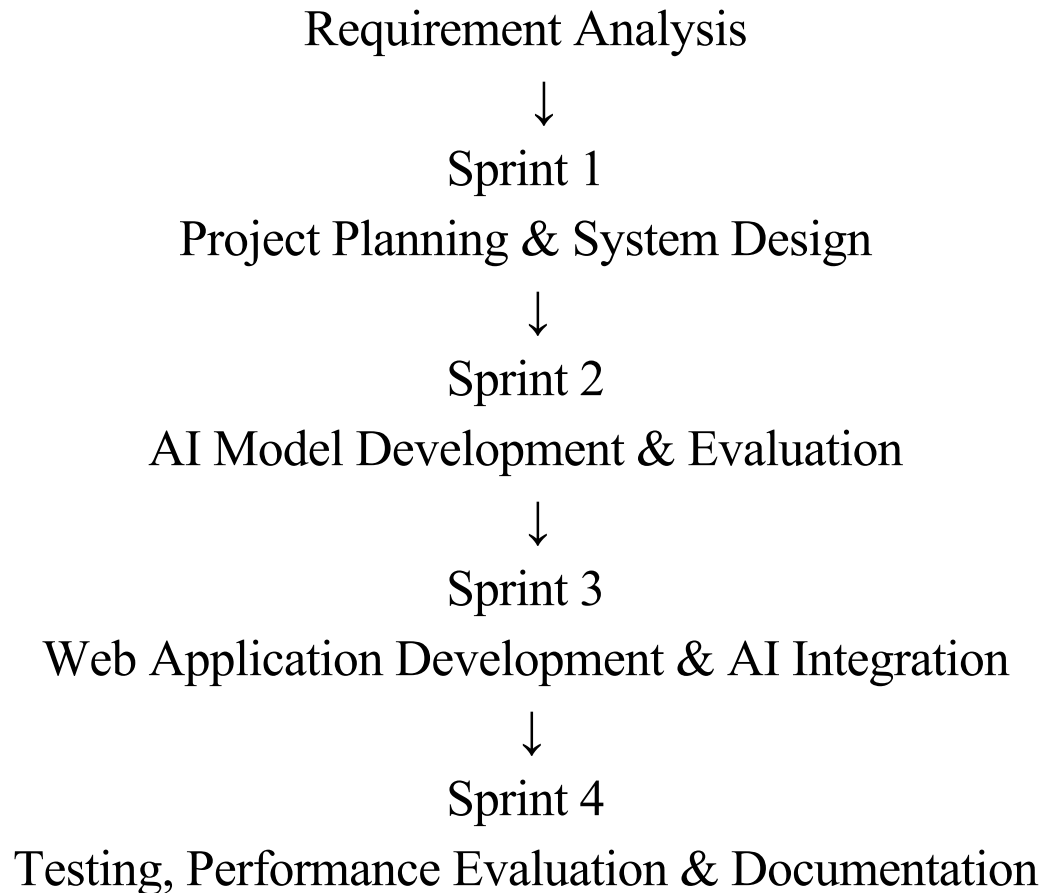
Methodology & Workflow



Methodology & Workflow

CompassIQ is developed using the Agile software development methodology. Agile divides the project into small development phases called sprints. Each sprint focuses on implementing and testing a specific set of features. This approach allows continuous improvement, regular feedback from the project guide, and easier management of project tasks

Methodology & Workflow



Methodology & Workflow

- Sprint 1 focuses on requirement analysis, database design, system architecture, UI design, and initial Flask/MySQL setup.
- Sprint 2 focuses on dataset preparation, NLP preprocessing, TF-IDF feature extraction, Logistic Regression models for category and priority prediction, and Cosine Similarity for retrieving similar historical tickets.
- Sprint 3 focuses on developing the Customer, Department, and Admin interfaces and integrating the trained AI models with the Flask backend and MySQL database.
- Sprint 4 focuses on functional testing, AI model evaluation, system performance testing, bug fixing, and project documentation.

MODULE DESCRIPTION

Customer Module : The Customer Module allows users to register and log in, submit support tickets, monitor ticket status, view responses from support departments, and provide feedback after resolution.

AI Intelligence Module: The AI Intelligence Module preprocesses customer complaints using NLP techniques and converts the text into TF-IDF feature vectors. Logistic Regression models predict the ticket category and priority. Cosine Similarity is used to retrieve the Top-3 similar historical complaints to assist support staff

MODULE DESCRIPTION

Department Module : The Department Module provides separate dashboards for Technical Support, Billing, Account Support, and General Inquiry. Tickets are automatically routed to the appropriate department based on the AI-predicted category. Department staff can view AI predictions, review similar historical tickets, respond to customers, update ticket status, and record final resolution details.

Administrator Module : The Administrator Module manages users and departments and provides overall system analytics. Administrators can monitor ticket statistics, category and priority distribution, department performance, and AI model performance.

THANK YOU